

Homework 5
COMS 3310
Due Wednesday April 15th, 11:59pm

No late work accepted.

Please read the HW guidelines (see next page). Failure to follow the guidelines will result in a lower grade.

1. (30 points) Design a NPDA for each of the following languages using JFLAP, using the indicated filenames, and submit them all to **Homework 5 PDAs** on Gradescope. The autograder for this part of the Gradescope assignment will be available latest by Thursday evening.

Please click on <https://www.jflap.org/tutorial/> and click on **Pushdown Automaton** and then **Construct and Run** for guidance on how to type in a PDA. When asked, choose **Single Character Input**. If asked, choose to use final states rather than the use of an empty stack when creating the PDA. Note that JFLAP uses Z as $\perp$ and puts this on the stack for you for free.

- **hw5q1L1.jff**: Let $|x|_a$ denote the number of occurrences of the symbol a in the string x . Design a NPDA for the following language over $\Sigma = \{a, b\}$.

$$L_1 = \{x \mid |x|_a = |x|_b\}$$

- **hw5q1L2.jff**:

$$L_2 = \{a^i b^j c^k \mid i = j \text{ or } j = 2k\}$$

- **hw5q1L3.jff**: L_3 over $\Sigma = \{a, b\}$ which contains only palindromes whose length is odd.
- **hw5q1L4.jff**: L_4 over $\Sigma = \{a, b\}$ which contains only palindromes whose length (modulo 6) is 2. That is, palindromes of length 2, 8, 14, ...

2. (20 points) This is an exercise in converting a CFG into CNF in a particular order. Design four CFGs in JFLAP for the language L_5 given by the following CFG:

$$\begin{aligned} S &\rightarrow TU \\ T &\rightarrow aTb \mid \epsilon \\ U &\rightarrow cUd \mid \epsilon \end{aligned}$$

First, note that this grammar already satisfies the START rule, that the start variable does not occur on the right side of any production (5 points each).

- (a) **hw5q2term.jff** Should satisfy the TERM rule: All terminal symbols should only occur on the right side of productions if the length of the right side is 1.
- (b) **hw5q2bin.jff** Should satisfy the previous rule, and the BIN rule: There should be no productions with a right hand side of length longer than 2.
- (c) **hw5q2del.jff** Should satisfy all the rules above and the DEL rule: There should be no productions with ϵ on the right side, except possibly the rule where the start variable yields ϵ .
- (d) **hw5q2unit.jff** Should satisfy all the rules above and the UNIT rule: There should be no productions where the right side is a single variable.

Please refer to https://en.wikipedia.org/wiki/Chomsky_normal_form or your notes taken from lecture, if you need more information about these rules.

Submit these grammars to **Homework 5 CNF** on Gradescope. Note that this is **not** the same as the assignment where the PDAs are submitted. The autograder for this part of the Gradescope assignment will be available latest by Thursday evening.

3. Use the Context-Free Pumping Lemma to prove that the following languages are not context-free. For this problem and the next, submit the handwritten and scanned answers on Gradescope **Homework 5 PDFs**.
- $L_6 = \{xx \mid x \in \{a,b\}^*\}$. Please start this on a new page.
 - (a) (5 points) First, given a pumping length k , try using a string $s = a^kba^kb$, and comment (in one or two sentences max) on why the proof fails.
 - (b) (15 points) Then choose a string that works and complete the proof.
 - (20 points) $L_7 = \{a^n b^{n^2} \mid n \geq 1\}$. This includes strings like ab , a^2b^4 , a^3b^9 . Please start this subproblem on a new page.

4. (10 points) Use the CYK algorithm (the dynamic programming algorithm presented in class) to determine whether the context-free grammar below generates the string *aabaa*. Please **print this page** and fill in the provided table, and include that with the rest of your answers when scanning your handwritten work.

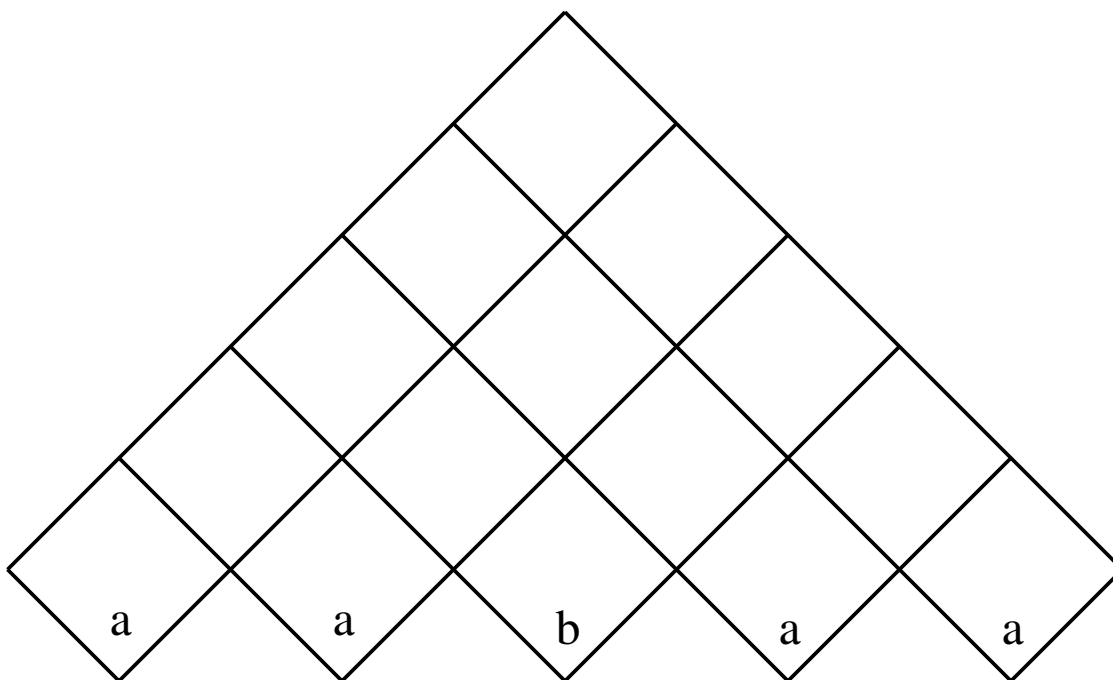
$$\begin{aligned}
 S &\rightarrow TA \mid BA \mid AB \mid b \\
 A &\rightarrow AC \mid a \\
 T &\rightarrow AB \\
 B &\rightarrow b \\
 C &\rightarrow a
 \end{aligned}$$


Figure 1: CYK Dynamic Programming Table

Guidelines:

- Please submit your HW via Gradescope.
- You must handwrite your solutions.
- If your handwriting is not legible, it will not be graded.
- Start answer to each question on a separate page.
- You are allowed to collaborate with *others*, while doing your homework.

- Definition of *others*: Students who are taking the course this semester.
- You can consult GenAI tools.
- However, you should write the final solutions alone, without consulting anyone. Your writing should demonstrate that you understand the proofs completely.
- For each problem, you should acknowledge all the individuals you consulted and whether you used any GenAI resources. This will not affect your grade. Failure to acknowledge is considered *academic dishonesty*, and it will affect your grade.
- When proofs are required, you should make them both clear and rigorous. Do not hand wave.
- Any regrading requests should be made within one week of returning the homework. No requests will be entertained after this one week period.